

- **What does the tr command do?**
 - Translates, deletes, or squeezes characters.
- **What does tr 'a-z' 'A-Z' do?**
 - Converts lowercase letters to uppercase.
- **What is the difference between using echo "text" | tr ... and using tr ... < file?**
 - Pipe uses command output as input.
 - `< file` reads directly from a file.
- **Did tr change the original file, or did it write output somewhere else?**
 - It only wrote output to stdout unless redirected.
- **What did tr '{}' '()' do?**
 - Replaced { with (and } with).
- **Why does tr work character-by-character instead of word-by-word?**
 - It is designed for single-character translation.
- **What happened to {?**
 - Changed to (.
- **What happened to }?**
 - Changed to).
- **What does tr -s do?**
 - Squeezes repeated characters into one.
- **What happened to the repeated spaces?**
 - Multiple spaces became a single space.
- **Why would squeezing repeated spaces be useful before processing a file?**
 - Makes formatting consistent for parsing.
- **What does tr -d do?**
 - Deletes characters.
- **What does tr -d '0-9' delete?**
 - All digits.
- **What does [:punct:] represent?**
 - All punctuation characters.
- **Why is tr -d useful for cleaning text?**
 - Removes unwanted characters quickly.
- **What does the -c option do in tr?**
 - Complements the character set (matches everything except).
- **What does tr -cd '0-9' mean?**
 - Keeps only digits.
- **Why did we include \n when keeping only letters and newlines?**
 - To preserve line breaks.
- **What is a regular expression?**
 - A pattern used for matching text.
- **What is a literal character match?**

- Matching exact characters.
- **Why did grep 'UNIX Tools' match some lines but not others?**
 - Only lines containing that exact text matched.
- **Can a regex match more than one place in the same line?**
 - Yes.
- **What does . mean in a regular expression?**
 - Any single character.
- **What did grep -o 'o.' print?**
 - Every match starting with o plus the next character.
- **What does the -o option do in grep?**
 - Prints only matching text.
- **Why can . be dangerous if you meant to search for an actual period?**
 - It matches any character unless escaped.
- **What is a character class?**
 - A set of characters inside brackets.
- **What does [aeiou] match?**
 - Any vowel.
- **What does [0-9] match?**
 - Any digit.
- **What does b[eor]at match?**
 - beat, boat, or brat.
- **Why does [eor] match only one character?**
 - Character classes match a single character.
- **What does [^...] mean?**
 - Negated character class.
- **What does b[^eo]at match?**
 - Words like brat but not beat or boat.
- **Why does b[^eo]at match brat but not beat or boat?**
 - e and o are excluded.
- **What is the difference between [abc] and [^abc]?**
 - [abc] matches listed chars.
 - [^abc] matches everything except them.
- **What does [:alpha:] match?**
 - Letters.
- **What does [:digit:] match?**
 - Digits.
- **What does [:alnum:] match?**
 - Letters and digits.
- **What does [:punct:] match?**
 - Punctuation symbols.

- **Why are named character classes useful for portability?**
 - Work consistently across locales/systems.
- **What does `^` mean in a regular expression?**
 - Start of line.
- **What does `$` mean in a regular expression?**
 - End of line.
- **What does `^$` match?**
 - Empty lines.
- **Why did we use `.$` instead of just `.$` to match lines ending with a period?**
 - `\.` matches a literal period.
- **What does `*` mean in a regular expression?**
 - Zero or more repetitions.
- **What does `ya*y` match?**
 - `yy`, `yay`, `yaay`, etc.
- *Why can a match empty text?**
 - Because zero repetitions are allowed.
- *What does `.` usually mean?**
 - Any number of any characters.
- **What does `{n}` mean?**
 - Exactly `n` repetitions.
- **What does `{n,}` mean?**
 - At least `n` repetitions.
- **What does `{n,m}` mean?**
 - Between `n` and `m` repetitions.
- **What does `a{2,}` match?**
 - Two or more `as`.
- **Why is `egrep` or `grep -E` easier for repetition ranges?**
 - Extended regex avoids escaping braces.
- **What is a subexpression?**
 - A grouped part of a regex in parentheses.
- **What does `(abc){2,3}` match?**
 - `abcabc` or `abcabcabc`.
- **What does the `|` symbol mean in extended regular expressions?**
 - OR operator.
- **Why do we use parentheses when we want repetition to apply to a group?**
 - Groups characters together as one unit.
- *Why does `[a-zA-Z_][a-zA-Z_0-9]*` work for many C-style variable names?**
 - Starts with letter/underscore, followed by letters, digits, or underscores.
- **Why does the dollar amount regex use `$`?**
 - Escapes `$` so it's treated literally.

- **What does `(.[0-9][0-9])?` mean?**
 - Optional decimal with two digits.
- **Why did the time regex match 9:30 am and 12:45 pm but not 19:99 pm?**
 - Regex restricted valid hour/minute ranges.
- **Why did the HTML header regex match , , and but not ?**
 - Pattern allowed only certain header numbers.
- **Why should regex patterns usually be placed inside single quotes?**
 - Prevents shell interpretation.
- **What does `ab` mean as a regular expression?**
 - Zero or more `as` followed by zero or more `bs`.
- **What does `a*b*` search for?**
 - Literal text `a*b*`.
- **What does the backslash `\` do in a regex?**
 - Escapes special characters.
- **What does `grep` do?**
 - Searches text using patterns.
- **What does `fgrep` do?**
 - Searches fixed strings (no regex).
- **What does `grep -F` do?**
 - Same as `fgrep`.
- **What does `egrep` do?**
 - Uses extended regular expressions.
- **What does `grep -E` do?**
 - Same as `egrep`.
- **Why is `fgrep` useful when searching for characters like `,` `^`, `$`, `[`, and `]`?*
 - Treats them as literal characters.
- **What is the difference between basic regular expressions and extended regular expressions?**
 - Extended regex supports more operators without escapes.
- **What does `grep -i` do?**
 - Ignore case.
- **What does `grep -c` do?**
 - Count matches.
- **What does `grep -l` do?**
 - Show matching filenames only.
- **What does `grep -w` do?**
 - Match whole words only.
- **What does `grep -o` do?**
 - Print only matching text.
- **What does `grep -n` do?**

- Show line numbers.
- **What does `grep -v` do?**
 - Invert match.
- **What does `grep -R` do?**
 - Recursive search.
- **What does `grep -A2` do?**
 - Show 2 lines after match.
- **What does `grep -B2` do?**
 - Show 2 lines before match.
- **What does `grep -C2` do?**
 - Show 2 lines before and after.
- **Why are context lines useful when reading search results?**
 - Provide surrounding information.
- **What does `grep -f` do?**
 - Reads patterns from a file.
- **Why might a pattern file be useful?**
 - Easier to manage many patterns.
- **How many patterns did you place in `patterns.txt`?**
 - Depends on your lab/example file.
- **What does `sed` stand for?**
 - Stream editor.
- **What does `s/unix/linux/` do?**
 - Replaces first `unix` with `linux`.
- **Why did the first command replace only the first `unix` on each line?**
 - Default behavior replaces one match per line.
- **What does the `g` flag do?**
 - Replaces all matches on the line.
- **What does `s/unix/linux/2` do?**
 - Replaces second occurrence only.
- **What does `s/unix/linux/3g` do?**
 - Replaces from third occurrence onward.
- **Did `sed` change the original file?**
 - No, unless `-i` was used.
- **What does `sed -n` do?**
 - Suppresses automatic printing.
- **What does the `/p` flag do?**
 - Prints matching lines.
- **Why did the command without `-n` print some lines twice?**
 - Default print plus explicit `/p`.
- **Why is `sed -n '/pattern/p'` similar to `grep 'pattern'`?**

- Both print matching lines.
- **What does `3s/unix/linux/g` do?**
 - Substitute only on line 3.
- **What does `1,3s/unix/linux/g` do?**
 - Substitute on lines 1–3.
- **What does `2,$s/unix/linux/g` do?**
 - Substitute from line 2 to end.
- **What does `$` mean in a sed address?**
 - Last line.
- **What does sed `'4d'` do?**
 - Deletes line 4.
- **What does sed `'10,12d'` do?**
 - Deletes lines 10–12.
- **What does sed `'/cow/d'` do?**
 - Deletes lines containing `cow`.
- **What does `1,5!d` mean?**
 - Keep only lines 1–5.
- **What does `!` mean in a sed command?**
 - Negates the match/address.
- **What does `/black/!s/cow/horse/` do?**
 - Replace `cow` with `horse` except on lines with `black`.
- **Which line stayed the same?**
 - The line containing `black`.
- **Which lines changed?**
 - Lines without `black`.
- **Why is `!` useful in sed?**
 - Applies commands to nonmatching lines.
- **What does the `i` command do in sed?**
 - Inserts text before a line.
- **What does the `a` command do in sed?**
 - Appends text after a line.
- **What does the `c` command do in sed?**
 - Changes/replaces a line.
- **Why do append, insert, and change commands require a backslash and a new line?**
 - Sed expects following text on next line.
- **What does the sed transform command `y` do?**
 - Character-by-character translation.
- **How is sed `'y/abc/xyz/'` similar to `tr 'abc' 'xyz'`?**
 - Both translate characters.
- **Why must the first and second character sets usually have the same length?**

- Each source character maps to one replacement.
- **What does sed '5q' do?**
 - Quits after line 5.
- **How is sed '5q' file similar to head -n 5 file?**
 - Both show first 5 lines.
- **Why can q save time on very large files?**
 - Stops reading early.
- **What is a sed script file?**
 - File containing multiple sed commands.
- **What does sed -f scripts/problem3_cleanup.sed do?**
 - Runs sed commands from the script file.
- **What did the script change?**
 - Depends on script contents.
- **What did the script delete?**
 - Depends on delete commands in script.
- **What did the script insert or append?**
 - Depends on insert/append commands used.
- **What did the report pipeline search for?**
 - Specific regex matches in text.
- **Why did we use egrep -o?**
 - Printed only regex matches.
- **Why did we pipe the output into tee?**
 - Saved output while displaying it.
- **What was saved into problem3_regex_report.txt?**
 - Regex match results/report output.
- **What does the script do?**
 - Automates regex/search processing tasks.
- **Why did the script use variables named input and report?**
 - Makes paths reusable and readable.
- **Why did we use chmod +x?**
 - Makes script executable.
- **Why is it useful to place repeated regex commands into a script?**
 - Saves time and ensures consistency.